MILITARY SPECIFICATION

PUMP ASSEMBLIES; FUEL, ELECTRICALLY DRIVEN, AIRCRAFT HEATER

This specification has been approved by the Bureau of Aeronautics, Department of the navy.

1. SCOPE

1.1 Scope. - This specification covers the requirements for procurement of one type of l lectrlcaUy driven, aircraft heater fuel pumps.

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS AND DRAWINGS

2.1 General. - The following specifications, publications and drawings of the issue in l ffect on the date of invitation for bids, form l part of this specifications

2.2 Specification. -

Federal Specification. -

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military Specifications . -

MIL-F-5572	Fuel; Aircraft Reciprocating Engine
MIL-F-5624	Fuel; Aircraft Turbine and Jot Engine, Grade JP-3

Air Force-Navy Aeronautical Specifications. -

AN-QQ-A-696	Anodic-Films; Corrosion-Protective (for) Aluminum Alloys
AN-C-150	Closure; Tubing Protective
AN-VV-C-576	Compound; Corrosion-Preventive, Aircraft Engine
AN-TT-E-501	Enamel ; Heat-Resisting, Glyceryl-Phthalate, Black
AN-F-42	Fluid; Hydrocarbon Standard Test
AN-M-40	Motors; Aircraft Direct Current, 24-Volt System
AN-P-32	Plating; Zinc
AN-P-38	Plates; Information and Identification
AN-P-62	Packaging of Lightweight . Aircraft Accessories
AN-P-363	Pipe Threads; Taper, Aeronaut ical National Form, Symbol ANPT
AN-P-656	Primer; Zinc Chromate
AN-S-126	Screw Threads; ; Standard, Aircraft
AN-W-22	Wire; Zinc Coated, Soft Steel

2.3 Other Publications. -

Bureau of Supplies and Accounts Publication. -

Navy Shipment Marking Handbook

2.4 Drawing. -

Air Force-Navy Aeronautical Design Standard Drawings. -

AND10049	Bosses - Straight Internal Thread Engine
AND10050	Bosses - Standard Dimensions for Gasket Seal Straight Thread

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2.5 Availability of Specifications, Publications, and Drawings. -

2.5.1 When requesting specifications, publication and drawings, refer to both title and number.

2.5.2 Copies of this specification and applicable specifications may be obtained upon application to the Commanding Officer, U. S. Naval Air Station, Johnsville, Pennyslvania, Attention: ion: Technical Records Division.

2.5.3 Copies of the Navy Shipment Marking Handbooks may be obtained upon application to the Bureau of Supplies and Accounts, Department of the Navy, Washington 25, D. C.

3. REQUIREMENTS

3.1 Pump Assembly. - .- The pump assembly shall be considered herein to consist of a pump-motor unit, radio noise filter, pressure relief valve, and fuel filter (if required by the pump manufacturer) as integral or separate parts.

3.2 Materials. - Materials used in the manufacture of fuel pump assemblies shall be of high quality, suitable for the purpose, and shall conform to applicable Government specifications. Materials conforming to the contractor's specifications may be used provided the specifications are released by the Government and contain provisions for adequate tests. The use of the contractor's specifications will net constitute waiver of Government. inspection.

3.2.1 Metals.- All metals used in the construction of the pump assembly shall be of a corrosion resisting type or shall be suitably protected to resist corrosion during the normal life of the pump assembly. The use of dissimilar metals especially brass, copper or steel in contact with aluminum or aluminum alloy shall be avoided whenever practicable.

3.2.2 Castings.- Castings shall be high quality, clean, sound, and free from blow holes, porosity, cracks, and other defects.

3.2.3 Threaded Parts. -

3.2.3.2 Screw Threads. - Screw threads shall be in accordance with Specification AN-S-126.

3.2.3.2 Pipe Threads. - Pipe threads shall be in accordance with Specification AN-P-363.

3.2.3.3 Locking wire. - All internal and external threaded parts, except drain plugs, shall be positively locked by means of wire conforming to Specification AN-W-22.

3 3.2.4 Standard Parts . - AN and JAN standard parts hsall be used whenever they are suitable for the purpose, and shall be identified on the drawing by their part number. Commercial utility parts such as screws, bolts, nuts, cotter pins, l tc., may be used provided they develop suitable properties and are replaceable by the AN or HAN standard parts Without alteration and provided the corresponding AN or JAN part numbers are referenced in the drawings and parts list. In applications for which no suitable corresponding AN or JAN part is in effect on date of invitation for bids, commercial parts may be used provided they conform to all the requirements of this specification.

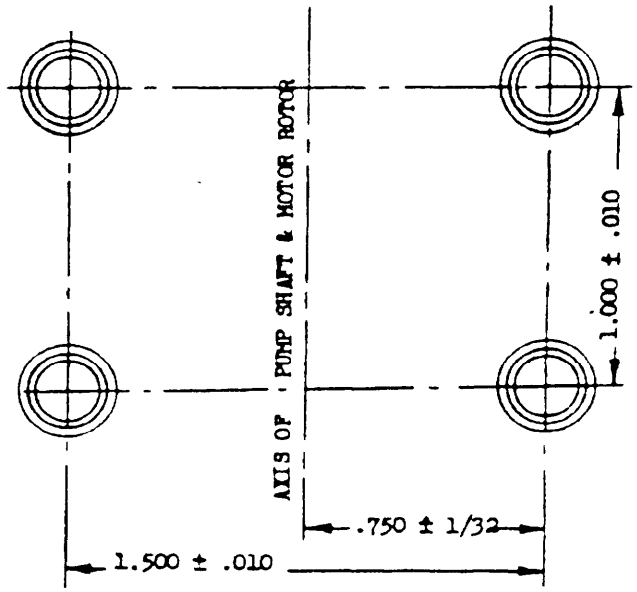
3.3 Design and Construction .- The pump assembly shall be constructed to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installations, and service use.

3.3.1 Dimensions. - The motor driven pump, excluding radio noise filter, pressure relief valve and fuel filter shall fit into a maximum space 7-5/8 x 3-3/8 x 3-1/8 inches.

3.3.1.1 Mounting. - The pump assembly shall be provided with bosses having four mounting holes. The holes shall be located and threaded as shown in Figure 1.

3.3.2 Weight. - The dry weight of the pump assembly shall not exceed 4.1 pounds.

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TAP DRILL DEPTH .438, C SINK 110° x .250, TAP
.250 - 20 NC-3, DEPTH .375 PERFECT THD. .312
4 HOLES AS SHOWN

FIGURE 1. PUMP MOUNTING BOSSES

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3.3.3 Power Consumption. - The electric power consumption for any condition of continuous operation shall not exceed 54 watts.

3.3.4 Ports. - Inlet and outlet ports shall conform to Drawing AND10049 or AND10050 for 1/4 inch outside diameter tubing.

3.3.5 Seals. - The pump assembly shall be provided with such fluid seals that no fuel leakage will occur.

3.3.5.1 Drains. - A sufficient number of drains shall be provided in the pump-motor unit to prevent fuel iron leaking into the motor with any normal pump position. The drains shall terminate on the outside of the prep-motor unit in bosses conforming to Drawing AND10049 or AND10050 for 1/4 inch outside diameter tubing. The bosses shall be equipped with plugs.

3.3.6 Relief Valves.- The relief valve shall be preferably of the balanced type. Other suitable arrangements whereby the pump discharge pressure will not be appreciably affected by variations in pump inlet pressure will be given consideration.

3.3.6.1 Pressure Adjustment. - The relief valve shall be so constructed that tthe fuel discharge pressure can be varied by some suitable spring adjusting screw arrangement which can be securely locked at any desired pressure setting within its range. There shall be a suitable adjustment stop provided to limit the maximum discharge pressure so that with 25.5 volts supplied to the motor and with zero discharge rate there shall be no damage to the motor or any other part of the pump assembly .

3.3.7 Motor. - The motor shall be in conformance with Specification AN-M-40 as specified for a continuous duty, explosion proof motor with an external filter. The connecting flanges of the pump and the motor may be as designed by the manufacturer.

3.3.8 Finish. -

3.3.8.1 Anodizing. - All aluminum alloy parts shall be anodized in accordance with Specificatoin AN-QQ-A-696.

3.3.8.2 Paint Finish. - All external surfaces except corrosion resisting steel, brass, cop-per, or bronze parts, working surfaces, threads, cadmium plated parts, anodized aluminum alloy, and zinc plated parts, shall be finished with one coat of primer conforming to Specification AN-P-656 and two coats of black enamel conforming to Specification AN-TT-E-501

3.3.8.3 Plating. - Non-rubbing steel parts in contact with aluminum or aluminum alloy shall be cadmium plated in accordance with Specification QQ-P-416 or zinc plated in accordance with Specification AN-F-32.

3.4 Performance. - The pump assembly shall satisfy all the performance requirements when subjected to the tests described in section 4.

3.5 Interchangeability. - All parts having the same manufacturer's s part number shall be directly and completely interchangeable with each other with respect. to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.6 Marking. -

3.6.1 Identification Marking. - Parts and subassemblies requiring replacement during normal maintenance shall be marked with the manufacturer' a part number and the manufacturer's name or trademark where practicable.

3.6.2 Nameplate Data. - The pump-motor unitet or an attached plate conforming to Specification AN-P-38 shall be permanently and legibly marked to give the following information:

PUMP - FUEL, ELECTRICALLY DRIVEN, AIRCRAFT HEATER
Specification MIL-P-6992 (Aer)
Manufacturer's Part No.

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Manufacturer's Serial No.
Contract No.
stock No.
Manufacturer's Name or Trademark

3.6.3 Ports. - The inlet and outlet ports shall be clearly and permanently identified.

3.6.4 Direction of Rotation. - An arrow shall be permanently marked on the pump to indicate the direction of rotation.

3.7 Workmanship. - Workmanship and finish on all parts shall be in accordance with high grade aircraft manufacturing practice.

4. SAMPLING, INSPECTOIN, AND TEST PROCEDURES

4.1 General Instructions. -

4.1.1 Previous Approval. - Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.1.2 Responsibility for Tests. - The cent tractor shall furnish all samples and shall be responsible for accomplishing the tests specifed herein.

4.1.3 Supervision of Tests. - When Inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government Inspector.

4.1.4 Laboratory Facilities. - Contractors not having laboratory facilities satisfactory to the Inspector shall engage the services of a commercial testing laboratory acceptable to the Inspector.

4.2 Sampling. -

4.2.1 Lot. - A lot shall consist of 50 or less pump assemblies of the same part number submitted for acceptance at the same time and place.

4.2.2 Representative Samples. - The Inspector shall select one pump assembly from each lot submitted for acceptance under contract for subjection to the tests of representative samples.

4.2.3 Qualification Test Samples. - Qualification test samples shall consist of two pump assemblies of each manufacturer's part number upon which qualification is desired. The units shall be forwarded to the Supply Officer, Naval Air Material Center, Philadelphia 12, Pennsylvania, marked "Attention: Superintendent, Aeronanutical Engine Laboratory, " plainly identified by se curely attached durable tags marked with the following information:

Sample for Qualification Tests
PUMP ASSEMBLES; FUEL, ELECTRICALLY DRIVEN, AIRCRAFT HEATER
Manufacturer's Part Number
Name of Manufacturer
Submitted by (name) (date) for qualification test in accordance with the requirements of the Specification MIL-P-6992(Aer) under authroization (reference authorizing letter)

4.2.3.1 Test Report. - Prior to delivery of the qualification test samples, a similar unit shall be tested by the manufacturer, in accordance with this specification. A complete test report showing results o fmanufacturer's tests and including calibration durves shall be forwarded with the qualification test sample.

4.2.3.2 Drawings. - A complete set of clear detail and assembly prints and materials list shall also be forwarded with the test samples.

4.3 Inspection. - Each pump assembly shall be carefully examined to determine conformance with the requirements of this specification not covered by tests.

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4.4 Tests. -

4.4.1 Individual Tests. - Each pump assembly submitted for acceptance shall be subjected to a break-in run and the following tests:

Body Leakage
Shaft Seal Leakage
Maximum Flow
Variable Voltage
Dry Inlet Suction

In addition, the pump assembly shall be subjected to any other tests specified herein which the Inspector considers necessary to determine conformance with the requirements of this specification.

4.4.2 Tests of Representative Samples. - Each representative sample shall be subjected to the individual tests and to the tests described in paragraphs headed as follows:

Pressure Adjustment
Effect of Negative Inlet Pressure on Discharge
Effect of Positive Inlet Pressure on Discharge

Qualification Test. - The qualification teat of pump assemblies shall consist of all this specification. Tests will normally be conducated in the sequence listed:

Pump Assembly No. 1	Pump Assembly No. 2
Explosion Proof	Break-In Run
Etmak-in Run	Leakage
Leakage	Calibration
Calibration	Low Temperature Operation
Fuel Resistance	Leakage
Leakage	Calibration
Calibration	Altitude Operation
Contaminated Fuel	Leakage
Calibration	Calibration
	Sea Level Endurance
	Leakage
	Calibration
	Orientation
	Vibration
	Radio Interference

Qualification tests may, at the option of the Qualifying Agency be supplemented with tests under actual or simulated service conditions.

4.5 Test Conditions. -

4.5.1 Cleaning. - Prior to test all oil and grease or other corrosion resisting compounds shall be removed from the interior parts of the pump assembly.

4.5.2 Voltage. - Unless otherwise specified, tests shall be conducted with a pump motor potential of 28.5 volts direct current.

4.5.3 Fuel. - Unless otherwise specified, fuel in accordance with Specification MIL-F-5572 Grade 100/130 or Grade 115/145 shall be used.

4.5.4 Flowmeter and Gages. - The flowmeter and gages shall be accurate to within 1-1/2 percent. The flowmeter and gages shall be calibrated to determine their accuracy at such times as deemed necessary by the Inspector.

4.5.5 Temperature. - Unless otherwise specified, the ambient and fuel temperature shall be approximately 70°F.

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4.5.6 Altitude. - Unless otherwise specified , tests shall be conducted at sea level pressure.

4.5.7 Test Setup. - The pump assembly shall he mounted on a test stand and connected as shown in Figure 2.

4.5.8 Lubrication. - No pump assembly shall be lubricated during any or all of the tests specified herein.

4.5.9 Break-In Run. - Prior to test the pump shall be subjected to not. less than one and not more than five hours break-h run, pumping fuel with a pump motor potential of 28.5 volts direct current and a pump discharge pressure of from 0 to 5 psi. If the manufacturer so desires, the first half of this run may be conducted using kerosene.

4.6 Test Methods. -

4.6.1 Leakage. -

4.6.1.1 Body Leakage. - The pump, with relief valve and fuel filter, if required, properly attached and all outlets plugged shall be submerged in a noncorrosive liquid and an internal pressure of 45 psi shall be applied through the inlet port of the assembly. The assembly shall be rejected if leakage occurs other than through the shaft seals.

4.6.1.2 Shaft Seal Leakage. - Shaft seal leakage shall be determined with a static fluid pressure of 25 psi applied on the inlet port with all other parts plugged. . There shall be no evidence of leakage at the shaft seals tithe r during this test or during any other test except during the body leakage test described above.

4.6.2 Calibration. -

4.6.2.1 Maximum Flow. - The maximum fuel flow at discharge pressures ranging from 3 to 30 pounds per square inch and zero inlet suction shall be determined. For this s test the relief valve shall be blocked shut or removed. The flow between 5 and 25 pounds per square inch discharge pressure shall be at least 50 pounds per hour.

4.6.2.2 Pressure Adjustment. - With each of various pressure adjustment settings the fuel discharge flow shall be varied from minimum to maximum. A curve of discharge rate versus discharge pressure shall be plotted for each pressure adjustment setting. Discharge pressure shall be satisfactorily obtainable at any valve ni.thin the range of 5 to 25 pounds per square inch and at any flow between 1 and 30 pounds per hour.

4.6.2.3 Effects of Negative Inlet Pressure on Discharge. - With an initial setting of 25 psi discharge pressure and 30 pounds per hour flow at zero inlet suction the inlet suction shall be varied in increments of 1 inch of mercury. At 3 inches mercury inlet suction the flow shall not have changed by more than 5 percent of the flow at zero inlet suction with the original settings of the relief valve and throttle valve maintained .

4.6.2.4 Effects of Positive Inlet Pressure on Discharge. - With an initial setting of 25 psi discharge pressure and 30 pounds per hour flow at zero inlet pressure the inlet pressure shall be varied from 0 to 25 pounds per square inch. The maximum allowable change in flow shall be 5 percent with the original settings of the relief valve and throttel valve maintained.

4.6.2.5 Variable Voltage. - With and initial setting of 30 pounds per hour flow and 25 pounds per square inch discharge pressure at zero inlet suction the voltage shall be varied from 28.5+ 5 to 19 volts direct current . The maximum allowable decrease in flow shall be 10 and 30 percent at 22 and 19 volts, respectively with the origianl settings of the relief valve and throttle valve maintained.

4.6.2.6 Dry Inlet Suction. - The pump shall have a minimum dry Inlet suction of two inches of mercury. Compressed air shall be blown through the purer while running for a period of one minute in order to thoroughly dry it . The dry inlet suction shall be measured with a mercury manometer. Total dry running time shall not exceed two minutes.

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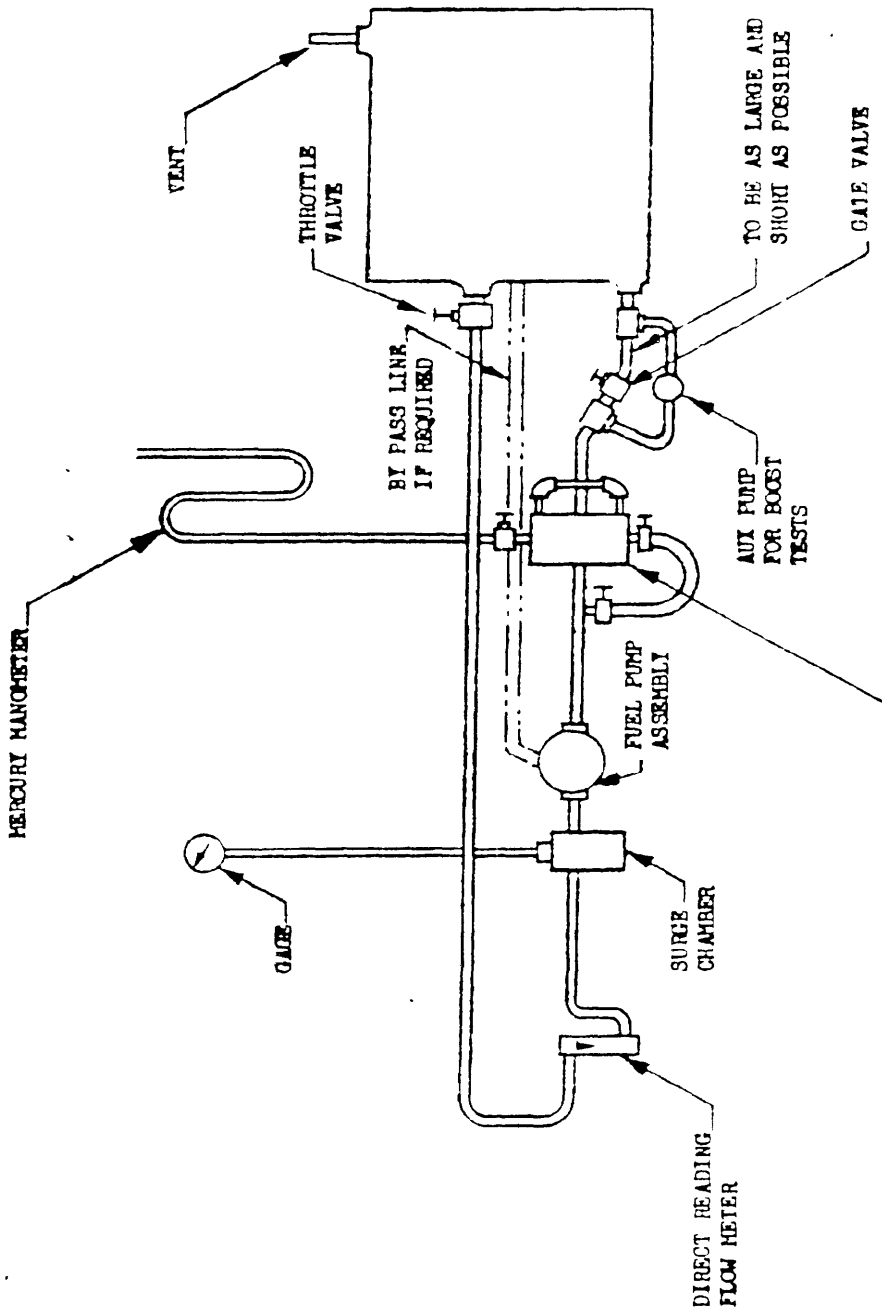


FIGURE 2. LABORATORY TEST SET-UP FOR FUEL PUMP ASSEMBLY

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4.6.3 Low Temperature Operation. - The pump assembly shall be capable of starting and sustaining operation at a minimum temperature of -65°F. Tests shall be conducated using test fluid Specification AN-F-42, Type I, at an inlet auction of 3 inches of mercury. The relief valve shall be set at 5 psi discharge pressure with 30 pounds per hour flow at room temperature. The pump assembly shall be stopped, drained of all fuel, and allowed to dry for a period of 24 hours at room temperature. The idle pump shall then be cooled to minus 65°F and maintained at this temperature for 24 hours in a dry condition. Fuel at a temperature of minus 65°F shall be introduced into the pump, and a teat run of approximately 5 to 10 minutes shall be made at minus 65°F. The fuel and the operating pump assembly shall then be allowed to warm up to room temperature, after which pressure and flow readings will agina be recorded. The pump assembly and fuel shall than be cooled to minus 65°F with the pump operating on a cycle of 10 hours ON and a minimum of 8 hours OFF for a total operating period of 20 hours. With the pump still operating, the temperature shall be raised to room temperature. The discharge pressure variation from the Initial setting shall not exceed minus 0.5 or plus 1.0 pounds per square inch at any time during the test. The above procedure shall be repeated with the relief valve set for 25 psi and with 30 pounds per hour flow. The discharge pressure variation from the initial setting shall not exceed minus 2.5 or plus 5.0 psi at any time during this test.

4.6.4 Altitude Operation. - Altitude tests shall be conducted on the pump assembly in an altitude chamber with a dry air atmosphere using fuel conforming to Specification MIL-F-5572 Grads 115/145. Pump inlet pressure shall be maintained at minus 1 inch of mercury gage.

4.6.4.1 Altitude Calibration.- A calibration shall be conducted with initial sea level settings of 25 psi discharge pressure and 30 pounds per hour flow. The pressure in the altitude chamber shall be decreased to simulate climb at the rate of 5000 feet per minute up to 50,000 feet. The pressure values as given in Table I shall be used in.this test. Satisfactory operatio shall be obtained at a minimum altitude of 15,000 feet with a discharge flow rate of at least 26 pounds per hour.

TABLE I ALTITUDE - PRESSURE

Altitude (1000 Feet)	Absolute Pressure (Inches of Mercury)
0	29.92
5	24.89
10	20.58
15	16.88
20	13.75
25	11.10
30	8.88
35	7.04
40	5.54
45	4.36
50	3.44

4.6.4.2 Altitude Endurance.- The pump assembly shall be operated for 250 hours with the relief valve set to give 25 psi gage discharge pressure at the minimum altitude chamber pressure necessary to maintain a discharge pressure fluctuation or decrease of approximately 5 psi. The pressure of the altitude chamber shall be decreased as the light ends of the fuel are boiled off in order to maintain the 5 psi fluctuntion or decrease during the complete test. The fuel used in the endurance test shall be changed every 50 hours. There shall be no adverse change in the performance of the pump assembly as a result of this test.

4.6.5 Sea-level Endurance. - The pump assembly shall be subjected to this endurance tact circulating fluids as specified for the various stages, of this test. Test fluids shall be maintained at a temperature of 80° to 100°F and changed after every 50 hours of operating time.

Stage 1: 150 hours of operation using test fluid Specificaion AN-F-42, Type IV, at 5 psi and 30 pounds par hour flow.

Stage 2: 150 hours Of operation using test fluid Specification AN-F-42, Type I, at 25 psi and 30 pounds per hour flow.

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Stage 3: 150 hours of operation using test fluid Specification AN-F-42, Type IV, at 25 psi and 6 pounds per hour flow ,

Stage 4: 150 hours of operation using test fluid Specification AN-F-42, Type I, at 5 psi and 6 pounds per hour flow.

Stage 5: 75 hours of operation using Specification MIL-F-5624 fuel at 25 psi and 30 pounds per hour flow.

Stage 6: 75 hours Of operation using Specification MIL-F-5624 fuel at 5 psi and b pounds per hour flow.

There shall be no failure of parts, no evidence of excessive wear, and no Averse change of more than 10 percent in any of the performance values as a result of all the tests up to and including this test.

4.6.6 Orientation. - With the relief valve set for 25 psi and 30 pounds per hour flow, the pump assembly shall operate satisfactorily for a period of at least 10 minutes in each of the following positions:

(a) 90• similated dive of the pump assembly from a horizontal position.

(b) 90″ simulated climb of the pump assembly from a horizontal position.

(c) Horizontal position and rotated about its longitudinal axis at 90° intervals.

4.6.7 Vibration. - The pump assembly shall be subjected to simple harmonic or circular motion applied to the mounting and the amplitude shall be 0.030 inch (maximum total excursion of 0.060 inch). The frequency of the vibration shall be established by cycle operation from approximately 10 cycles to 50 cycles and back to 10 cycles per second. The duration of one such cyclic operation shall be between one and five minutes. The pump assembly shall be vibrated continuously for two hours in the direction of each of the three major axes. The pump assembly shall be operated during one hour of each 2-hour period. The pump assembly shall function satisfactorily throughout the vibration range and at the completion of the vibration tests. No damage to the pump assembly shall result from the vibration tests.

4.6.8 Fuel Resistance. - The pump assembly shall be subjected to “Calibration” prior to the “Fuel Resistance” test. Using a minimum of two gallons of fuel, the pump assembly shall be operated in or immersed in Specificaion AN-F-42, Type IV, fluid at a temperature of 70° (158°F) for a period of 300 hours in such a manner as to insure complete contact of synthetic parts with the fluid as would be expected under service conditions. The pump assembly shall then be oven-dried with open ports at a temperature of 158°F for a period of four hours. Immediately thereafter the same pump assembly shall be operated in or immersed in Specification AN-F-U, Type I, fuel at a temperatuer of 70° (for a period of 300 hours. The pump assembly shall then be oven-dried, with open ports, at d temperature of 70°C (158°F) for a period of four hours. Prior to conductin this fuel resistance test and subsequent to each drying period, the pump assembly shall be tested in accordance with the paragraph headed ‘Leakage. s Prior to each drying period the fuel filter element shall be removed and rein-stalled ten times, using all original parts. After each reassembly the filter shall be tested for body leakage with 45 psi internal air pressure while submerged in a noncorrosive liquid. Any leakage as evidenced by bubbles shall be cause for rejection. The pump assembly with the original seals and gaskets, shall be maintained at a temperature of -65 to -70°F for 24 hours. During this time, the pump assembly shall be operated using Specification AN-F-42, Type I, fluid at a 25 pounds per square inch discharge pressure and 30 pounds per hour flow with an inlet pressure of 20 to 25 pounds per square inch. At the completion of this test the pump assembly shall be subjected to "Calibration" and there shall be no adverse change of mere than 10 percent in any of the performance values as determined by "Calibration" prior to the "Fuel Resistance" test.

4.6.9 Contaminated Fuel. - The pump assembly shall be operated st a 25 psi discharge pressure and 30 pounds par hour flow while pumping 20 gallons of fuel contaminated as specified in Table IX.

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TABLE II

Contaminant for Fuel

Contaminant Description	Particle Size	Quantity
Sharp silica sand	Smaller than 30 mesh Larger than 50 mesh	.2 grams
Sharp silica sand	Smaller than 50 mesh Larger than 100 mesh	.2 grams
Prepared dust AC Spark Plug Co. Part No. 1543637 or equal	0-5 microns 12±2% present 5-10 microns 12±3% present 10-20 microns 14±3% present 20-40 microns 23±3% present 40-80 microns 30±3% present 80-200 microns 9±3% present	1.6 grams

contaminated fuel, which is well agitated to maintain a uniform mixture in the supply tank, shall be pumped throught a recirculating system for a total operating time of 10 hours. There shall be no adverse change in the performance of the pump assembly as a result of this test.

4.6.10 Radio Interference. -

4.6.10.1 Conducted Radio Interference. - The equipment and test method used shall conform to Specification JAN-I-225. The conducted radio interference produced by operation of the pump motor at both rated torque and minimum operating load and measured between each terminal and ground or between any two terminals shall not exceed 200 microvolts in the frequency ramge 0.15 to 0.20 megacycles and 50 microvolts in the frequency range 0.20 to 20.0 megacycles. Measurement shall not be made between an open circuit and any other point.

4.6.10.2 Radiated Radio Interference. - Radiated interference produced by operation of the pump motor at both rated torque and minimum specified operating load shall not exceed the following values:

ALLOWABLE LIMIT IN EQUIVALENT MICROVOLTS	FREQUENCY RANGE IN MC SEC.	CALIBRATED RECEIVERS
3	0.2 to 0.5	B-C-348Q
1	1.5 to 18	B-C-348Q
Background Level of Receiver	100 to 156	B-C-639

Measurements shall be made with calibrated receivers as specified using an electrostatically shielded loop probe antanna conforming with Drawing AN3065. The probe shall be oriented and positioned with respect to the equipment and cabling so as to obtain maximum interference indication. An audio output meter shall be used, either the Weston No. 687 voltmeter or an equivalent voltmeter having not less than 1000 ohms per volt. One set of headphones, ANB-H-1 or ANB-H-1a, shall be used in parallel with the output meter for monitoting purposes and to match the output impedance of the receiver. All receiver controls, not internal and external, shall be adjsuted to obtain maximum output and MCM reception employed. Squelch and limiter circuits shall be made inoperative.

4.6.11 Explosion Proof. - The explosion proof test shall be conducted in a test chamber having a transparent window on one side and the opposite side opening closed by thin non-porous paper to permit instant pressure relief during explosion. The chamber shall be

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equipped with a suitable means for vaporizing fuel and maintaining a predetermined gasoline vapor-air mixture throughout the test chamber. Three drilled holes in the motor housing shall be tapped for insertion of l spark plug, a vapor line and a vent line . A pressure gage shall be connected in the vapor line to indicate when explosions occur in the motor housing. The motor and surrounding chamber shall be filled with the explosive mixture end the valves in the vapor and vent lines shall be closed. The motor shall be operated at approximately rated load for a period of 15 minutes. During this period the explosive mixture In the motor shall b. exploded (either by arcing within the motor or by means of the spark plug) and replaced for three explosions. At the end of the 15 minute period the mixture in the chamber shall be exploded to determine that an explosive mixture existed in the chamber. A total of 15 teats as described above shall be conducted on the motor with l xpbsive mixture having air-fuel ratios by weight, of 12.6:1 and 15.5:1, and other mixtures spaced between these limiting values. The pump motor shall confine all sparks and flame within the motor. Any explosion within the motor shall not ignite the explosive mixture surrounding the motor.

4.7 Rejections and Retest. - When any representative sample fails to meet the requirements of the tests of representative samples, the lot represented ted shall be rejected and returned at the contractor's expense. Any pump assembly failing to meet. the requirements of the individual tests shall be rejected and returned at the contractor's expense. Pumps which have been rejected may be replaced or repaired to correct the defects and resubmitted for tests. When this has been accomplished, all specified trots shall be repeated. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the original defects shall be furnished the Inspector. Unite rejected after retest shall not be resubmitted without the specific approval of the Bureau of Aeronautics.

5. PREPARATION FOR DELIVERY

5.1 Application of Requirements for Preparation for Delivery. - The requirements specified in section 5 apply only to direct purchases by or direct shipments to the Government.

5.2 Preservation and Packaging. - Each pump assembly shall be protected form corrosion or deterioration during shipment ant!,'or storage by flushing internally with Type II corrosin preventive compound conforming to Specification AN-VV-C-576 and openings sealed with closures conforming to Specification AN-C-150. Each pump assembly shall then be preserved and packaged in accordance with Specification AN-P-62, Group VII, Type II cent ainer.

5.3 Packing. - Unless otherwise specified by the Procuring Agency, pump assembly packages shall be overpacked in accordance with Specification AN-P-62.

5.4 Labeling and Marking. -

5.4.1 Interior Container. - Each interior container shall be marked with the following in information with the same size type or lettering in such a manner that the marking will not become damaged when the packages are opened:

PUMP ASSEMBLY ; FUEL , ELECTRICALLY DRIVEN , AIRCRAFT HEATER
Specification MIL-P-6992(Aer)
Stock Number _____
Quantity Contained _____
Manufacturer's Part Number _____
Manufacturer's Serial Number _____
Name of Contractor (and Name of Manufacturer if not the same)
Government Order Number (or Contract Number if Order Number is not assigned)
Date of Preservation

5.4.1.1 Precautionary Marking.- The following precautionary marking shall appear on two opposite sides of each interior package:

"FRAGILE
DELICATE EQUIPMENT
HANDLE WITH CARE"

5.4.2 Exterior Containers. - Shipments shall be marked in accordance with the requirements of the Navy Shipment Marking Handbook. In addition, each exterior container shall be marked as specified for interior packages, marked to indicate presence of preservation and method of packing, i.e., domestic or overseas.

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6. NOTES

6.1 Intended Use. - The 28 volts direct current motor driven fuel pump assemblies covered by this specification are intended for use on aircraft, to supply fuel to aircraft heaters.

6.2 Ordering Data. - Requisitions, contracts, and orders should indicate the quantity desired and state whether overseas packing is required (see section 5).

6.3 Qualification Test. - IN the procurement of materials requiring qualification (type or brand approval), the right is reserved to reject any bids on pump assemblies which have not been subjected to the retired teats and found satisfactory for inclusino in the Navy Department acceptable list of approval materials. The attention of the manufacturers is called to this provision, and they are urged to request authorization for tests of the pump assemblies which they props. to offer to the Navy under this specification in order that they may be l l&iblm to be awarded contracts or orders. Requests for authorization of tests, and for information as to the test fees involved should be addressed to the Bureau of Aeronautics, Navy Department, Washington 25, D. C. It is to be understood that the manufacturer shall pay all transportation charges to and from the point where tests are made. IN the case of failure of the sample or samples submitted, consideration will be given to the request of the manufacturar for additional tests only after it has been clearly shown that changes have been made in the product which the Government considers sufficient to warrant additional toots.

6.3.1 It is to be understtod that pump assemblies supplied under contract shall be identical in every respect to the samples tested and found satisfactory, except for changes previously approved by the Government. Any unapproved changes from the qualification sample shall constitute cause for rejection.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connectoin with a definitely related Government procurement operation, the United States Government the reby incure no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specification, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporatoin, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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